

Vasilis Kontonis

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Experience

Microsoft Research

Senior Researcher, Jul 2025 – present
New York City, NY, USA

University of Texas at Austin

IFML Postdoctoral Fellow Sep 2023 – Jul 2025
Austin, TX, USA

Google

Research Intern, May 2022 – Nov 2022
Mountain View, CA, USA

GRNET (Greek Research and Technology Network)

Systems and Services Engineer, 2016
Athens, Greece

Education

University of Wisconsin-Madison

Ph.D., Computer Science, 2018 – 2023.

Thesis: Learning From Imperfect Data: Noisy Labels, Truncation,
and Coarsening

Advisor: Christos Tzamos

National Technical University of Athens

Master of Engineering in Electrical & Computer Engineering

Thesis: Learning Powers of Poisson Binomial Distributions

Advisor: Dimitris Fotakis

Awards

Selected for Oral Presentation

Conference on Language Modeling (COLM) 2026

San Francisco, CA, USA

MEMENTO: Teaching LLMs to Manage Their Own Context

Outstanding Paper Award

International Conference on Machine Learning (ICML) 2025

Vancouver, CA

Train for the Worst, Plan for the Best

Understanding Token Ordering in Masked Diffusions

Best Paper Award

Conference on Learning Theory (COLT) 2024

Edmonton, CA

Smoothed Analysis for Learning Concepts with Low Intrinsic
Dimension

Research Interests	Machine Learning, Statistics, Theoretical Computer Science
Preprints	Scaling Discovery through Test-Time Communication Jongho Park, Vasilis Kontonis, Shivam Garg, Akshay Krishnamurthy, Dimitris Papailiopoulos <i>arXiv:2609.21032, September 2026</i>
Publications	Author names follow the order in the published paper. Preprints are listed separately. MENTO: Teaching LLMs to Manage Their Own Context Vasilis Kontonis, Yuchen Zeng, Shivam Garg, Lingjiao Chen, Hao Tang, Ziyang Wang, Ahmed Awadallah, Eric Horvitz, John Langford, Dimitris Papailiopoulos <i>Third Conference on Language Modeling</i> Selected for Oral Presentation COLM 2026 Wait, Wait, Wait... Why Do Reasoning Models Loop? Charilaos Pipis, Shivam Garg, Vasilis Kontonis, Vaishnavi Shrivastava, Akshay Krishnamurthy, Dimitris Papailiopoulos <i>43rd International Conference on Machine Learning</i> Spotlight Presentation ICML 2026 Train for the Worst, Plan for the Best: Understanding Token Ordering in Masked Diffusions J. Kim, K. Shah, V. Kontonis, S. M. Kakade, S. Chen <i>Proceedings of the 42nd International Conference on Machine Learning</i> Outstanding Paper Award ICML 2025 Learning General Gaussian Mixtures with Efficient Score Matching S. Chen, V. Kontonis, K. Shah <i>Proceedings of the 38th Annual Conference on Learning Theory</i> COLT 2025 Online Linear Classification with Massart Noise I. Diakonikolas, V. Kontonis, C. Tzamos, N. Zarfis <i>Proceedings of the 42nd International Conference on Machine Learning</i> ICML 2025

Oracle-Efficient Truncated Statistics

K. Karatapanis, V. Kontonis, C. Tzamos

Proceedings of the 13th International Conference on Learning Representations

ICLR 2025

Smoothed Analysis for Learning Concepts with Low Intrinsic Dimension

Gautam Chandrasekaran, Adam Klivans, Vasilis Kontonis, Raghu Meka, Konstantinos Stavropoulos

Proceedings of the 37th Annual Conference on Learning Theory

Best Paper Award

COLT 2024

Active Learning with Simple Questions

Vasilis Kontonis, Mingchen Ma, Christos Tzamos

Proceedings of the 37th Annual Conference on Learning Theory

COLT 2024

Agnostically Learning Multi-index Models with Queries

Ilias Diakonikolas, Daniel M. Kane, Vasilis Kontonis, Christos Tzamos, Nikos Zarifis

Proceedings of the 65th Annual Symposium on Foundations of Computer Science

FOCS 2024

Super Non-singular Decompositions of Polynomials and their Application to Robustly Learning Low-degree PTFs

Ilias Diakonikolas, Daniel M Kane, Vasilis Kontonis, Sihan Liu, Nikos Zarifis

Proceedings of the 56th Annual ACM Symposium on Theory of Computing

STOC 2024

Active Classification with Few Queries under Misspecification

Vasilis Kontonis, Mingchen Ma, Christos Tzamos

Proceedings of the 38th Conference on Neural Information Processing Systems

Spotlight Presentation

NeurIPS 2024

Learning Noisy Halfspaces with a Margin: Massart is no Harder than Random

Gautam Chandrasekaran, Vasilis Kontonis, Kostas Stavropoulos, Kevin Tian

Proceedings of the 38th Conference on Neural Information Processing Systems

Spotlight Presentation

NeurIPS 2024

Efficient Discrepancy Testing for Learning with Distribution Shift

Gautam Chandrasekaran, Adam Klivans, Vasilis Kontonis, Kostas Stavropoulos, Arsen Vasilyan

Proceedings of the 38th Conference on Neural Information Processing Systems

NeurIPS 2024

Optimizing Solution-Samplers for Combinatorial Problems: The Landscape of Policy Gradient Methods

Constantinos Caramanis, Dimitris Fotakis, Alkis Kalavasis, Vasilis Kontonis, Christos Tzamos

Proceedings of the 37th Conference on Neural Information Processing Systems

Selected for Oral Presentation

NeurIPS 2023

SLaM: Student Label Mixing for Distillation with Unlabeled Examples

Vasilis Kontonis, Fotis Iliopoulos, Khoa Trinh, Cenk Baykal, Gaurav Menghani, Erik Vee

Proceedings of the 37th Conference on Neural Information Processing Systems

NeurIPS 2023

The Gain from Ordering in Online Learning

Vasilis Kontonis, Mingchen Ma, Christos Tzamos

Proceedings of the 37th Conference on Neural Information Processing Systems

NeurIPS 2023

Efficient Testable Learning of Halfspaces with Adversarial Label Noise

Ilias Diakonikolas, Daniel M Kane, Vasilis Kontonis, Sihan Liu, Nikos Zarifis

Proceedings of the 37th Conference on Neural Information Processing Systems

NeurIPS 2023

Self Directed Linear Classification

Ilias Diakonikolas, Vasilis Kontonis, Christos Tzamos, Nikos Zarifis
Proceedings of the 36th Annual Conference on Learning Theory
COLT 2023

Opinion Dynamics with Limited Information

Dimitris Fotakis, Vardis Kandiros, Vasilis Kontonis, Stratis Skoulakis
Algorithmica
ALGORITHMICA 2023

Weighted Distillation with Unlabeled Examples

Fotis Iliopoulos, Vasilis Kontonis, Cenk Baykal, Gaurav Menghani, Khoa Trinh, Erik Vee
Proceedings of the 36th Conference on Neural Information Processing Systems
NeurIPS 2022

Linear Label Ranking with Bounded Noise

Dimitris Fotakis, Alvertos Kalavasis, Vasilis Kontonis, Christos Tzamos
Proceedings of the 36th Conference on Neural Information Processing Systems
Selected for Oral Presentation
NeurIPS 2022

Learning General Halfspaces with General Massart Noise under the Gaussian Distribution

Ilias Diakonikolas, Daniel M. Kane, Vasilis Kontonis, Christos Tzamos, Nikos Zarifis
Proceedings of the 54th Annual ACM Symposium on Theory of Computing
STOC 2022

Learning a Single Neuron with Adversarial Label Noise via Gradient Descent

Ilias Diakonikolas, Vasilis Kontonis, Christos Tzamos, Nikos Zarifis
Proceedings of the 35th Annual Conference on Learning Theory
COLT 2022

Learning General Halfspaces with Adversarial Label Noise via Online Gradient Descent

Ilias Diakonikolas, Vasilis Kontonis, Christos Tzamos, Nikos Zarifis
Proceedings of the 39th International Conference on Machine Learning
ICML 2022

A Statistical Taylor's Theorem and Extrapolation of Truncated Densities

Costis Daskalakis, Vasilis Kontonis, Christos Tzamos, Manolis Zampetakis

*Proceedings of the 34th Annual Conference on Learning Theory
COLT 2021*

Efficient Algorithms for Learning from Coarse Labels

Dimitris Fotakis, Alvertos Kalavasis, Vasilis Kontonis, Christos Tzamos

*Proceedings of the 34th Annual Conference on Learning Theory
COLT 2021*

Agnostic Proper Learning of Halfspaces under Gaussian Marginals

Ilias Diakonikolas, Daniel M. Kane, Vasilis Kontonis, Christos Tzamos, Nikos Zarifis

*Proceedings of the 34th Annual Conference on Learning Theory
COLT 2021*

Learning Online Algorithms with Distributional Advice

Ilias Diakonikolas, Vasilis Kontonis, Christos Tzamos, Ali Vakilian, Nikos Zarifis

*Proceedings of the 38th International Conference on Machine Learning
ICML 2021*

A Polynomial Time Algorithm For Learning Halfspaces with Tsybakov Noise

Ilias Diakonikolas, Daniel M. Kane, Vasilis Kontonis, Christos Tzamos, Nikos Zarifis

*Proceedings of the 53rd Annual ACM Symposium on Theory of Computing
STOC 2021*

Learning Halfspaces with Tsybakov Noise

Ilias Diakonikolas, Vasilis Kontonis, Christos Tzamos, Nikos Zarifis

Proceedings of the 53rd Annual ACM Symposium on Theory of Computing

STOC 2021 (Conference version merged with the above paper)

Non-Convex SGD Learns Halfspaces with Adversarial Label Noise

Ilias Diakonikolas, Vasilis Kontonis, Christos Tzamos, Nikos Zarifis

*Proceedings of the 34th Conference on Neural Information Processing Systems
NeurIPS 2020*

	Learning Halfspaces with Massart Noise Under Structured Distributions Ilias Diakonikolas, Vasilis Kontonis, Christos Tzamos, Nikos Zarifis <i>Proceedings of the 33rd Annual Conference on Learning Theory</i> COLT 2020 Algorithms and SQ lower bounds for PAC Learning One-Hidden-layer ReLU Networks Ilias Diakonikolas, Daniel M. Kane, Vasilis Kontonis, Nikos Zarifis <i>Proceedings of the 33rd Annual Conference on Learning Theory</i> COLT 2020 Efficient Truncated Statistics with Unknown Truncation Vasilis Kontonis, Christos Tzamos, Manolis Zampetakis <i>Proceedings of the 60th Annual IEEE Symposium on Foundations of Computer Science, FOCS 2019</i> Opinion Dynamics with Limited Information Dimitris Fotakis, Vardis Kandiros, Vasilis Kontonis, and Stratis Skoulakis <i>Proceedings of the 14th Conference on Web and Internet Economics</i> WINE 2018
Journal	Agnostically Learning Multi-Index Models with Queries Ilias Diakonikolas, Daniel Kane, Vasilis Kontonis, Christos Tzamos, Nikos Zarifis <i>SIAM Journal on Computing, Special Section FOCS 2024</i> FOCS24-435–FOCS24-486, July 2026
Teaching	Opinion Dynamics with Limited Information Dimitris Fotakis, Vardis Kandiros, Vasilis Kontonis, and Stratis Skoulakis <i>Algorithmica</i> 85, no. 12 (2023): 3855-3888 Spring 2023, National Technical University of Athens “Online Learning and Bandits” Instructors: Constantine Caramanis, Vasilis Kontonis Spring 2022, University of Wisconsin-Madison Teaching Assistant, “Introduction to Artificial Intelligence”, CS 540 Instructors: Sharon Li, Ilias Diakonikolas, Jerry Zhu Fall 2021, University of Wisconsin-Madison Teaching Assistant, “Introduction to Numerical Methods”, CS 412 Instructor: Amos Ron Spring 2021, University of Wisconsin-Madison Teaching Assistant, “Introduction to Algorithms”, CS 577 Instructors: Christos Tzamos, Marc Renault

Service

Fall 2020, University of Wisconsin-Madison

Teaching Assistant, “Programming II”, CS 300

Instructors: Mouna Ayari Ben Hadj Kacem, Laura Legault

Spring 2020, University of Wisconsin-Madison

Teaching Assistant, “Introduction to Algorithms”, CS 577

Instructor: Christos Tzamos

Spring 2019, University of Wisconsin-Madison

Teaching Assistant, “Introduction to Algorithms”, CS 577

Instructor: Dieter van Melkebeek

Fall 2018, University of Wisconsin-Madison

Teaching Assistant, “Introduction to Algorithms”, CS 577

Instructors: Christos Tzamos, Shuchi Chawla

Fall 2017, National Technical University of Athens

Teaching Assistant, “Algorithms and Complexity”

Instructor: Aris Pagourtzis

Fall 2016, National Technical University of Athens

Teaching Assistant, “Operating Systems”

Instructors: Nectarios Koziris, Georgios Goumas, Vangelis Koukis

Program Committees

ITCS 2025

Conference Reviewer

STOC 2025, FOCS 2024, STOC 2024, NeurIPS 2023, ICML 2022,
NeurIPS 2021, ICML 2021, EC 2021, STOC 2020, ALT 2020,
SODA 2019

ICML 2019, EC 2019, WINE 2019, MFCS 2018

Journal Reviewer

Theoretical Computer Science (TCS)

Talks

MEMENTO: Teaching LLMs to Manage Their Own Context

*Upcoming oral presentation, COLM 2026, San Francisco, CA
October 2026*

Learning General Gaussian Mixtures with Efficient Score Matching

2024, IFML/MPG Symposium, Simons Institute, UC Berkeley

Learning from Noisy Labels: Beyond Worst Case Models

*2024, UC Santa Cruz, Computer Science and Engineering
Colloquium*

*2024, New Directions in Machine Learning Theory Workshop,
BIRS, Canada*

Learning from Noisy Labels and Imperfect Teachers

2024, UC San Diego, EnCORE Presentation

2023, UT Austin, IFML Seminar

Smoothed Analysis for Learning Concepts with Low Intrinsic Dimension

2024, INFORMS Annual Meeting, Seattle, WA

2024, COLT, Best Paper Award Talk

2024, EnCORE Workshop on Computational vs Statistical Gaps in Learning and Optimization, UCLA

2024, Theory Seminar, USC

2024, Theory Seminar, University of Texas at Austin

Optimizing Solution-Samplers for Combinatorial Problems

2023, 37th Conference on Neural Information Processing Systems (Oral, NeurIPS 2023)

SLaM: Student-Label Mixing for Distillation with Unlabeled Examples

2023, 37th Conference on Neural Information Processing Systems (NeurIPS 2023)

Linear Label Ranking with Bounded Noise

2022, 36th Conference on Neural Information Processing Systems (NeurIPS 2022)

Learning General Halfspaces with General Massart Noise under the Gaussian Distribution

2022 54th ACM Symposium on Theory of Computing (STOC2022)

A Statistical Taylor's Theorem and Extrapolation of Truncated Densities

2021, 34th Annual Conference on Learning Theory (COLT 2021)

Agnostic Proper Learning of Halfspaces under Gaussian Marginals

2021, 34th Annual Conference on Learning Theory (COLT 2021)

Efficient Algorithms for Learning Halfspaces with Tsybakov Noise

2021, 53rd Symposium on Theory of Computing (STOC 2021)

Non-Convex SGD Learns Halfspaces with Adversarial Label Noise

2020, 34th Conference on Neural Information Processing Systems (NeurIPS 2020)

Learning Halfspaces with Massart Noise Under Structured Distributions

2020, 33rd Annual Conference on Learning Theory (COLT 2020)

Fellowships &
Grants

Truncated Statistics with Unknown Truncation

Symposium on Foundations of Computer Science (FOCS 2019)

Workshop on Algorithms for Learning and Economics (WALE 2019)

14th Athens Colloquium on Algorithms and Complexity (ACAC 2019)

Theory Study Group, University of Wisconsin-Madison

Learning Powers of Poisson Binomial Distributions

2017, ECCO Research Seminar, University of Liverpool

Bias-Variance Tradeoff , VC-Dimension

2017, Learning Theory Study Group, Corelab, NTUA

**Applications of LP, QP, SOCP and SDP. SDP Relaxations:
MaxCut. Vector Optimization, Duality**

2017, Convex Optimization Minicourse, Corelab, NTUA.

Bodossakis Fellowship 2022, Gerondellis Fellowship 2020

Eurobank Grant “The great moment for Education” 2010

Touramanoglou Grant 2010